5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

SGSMA
2026
Smart Grid Synchronized Measurements and Analytics

Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Date
June 1-4, 2026

Venue
Santiago, Chile

Motivation

Why frequency estimation in low-inertia IBR grids is hard

Field events define the benchmark stress model

2016	2017	2021	2021	2022
Blue Cut CA, Aug. 16 **1,178 MW PV** false near-instantaneous frequency trips during distorted fault waveforms *stress: distortion → frequency artifact*	**Canyon 2** CA, Oct. 9 **682 / 937 MW** solar reductions; erroneous frequency tripping was not the cause *stress: PLL, voltage response, recovery*	**Odessa I/II** TX faults, 2021/22 **1,112 / 1,711 MW** remote solar reductions after normally cleared faults *stress: phase jumps, controls, model mismatch*	**Kaua'i** Nov. 21 **18–20 Hz oscillation** near or beyond ordinary 30 fps phasor-rate observability *stress: oscillation and aliasing*	**CA BESS** Mar. 9 / Apr. 6 **BESS/IBR reductions** ac overcurrent/overvoltage, dc-bus, unbalance, data-quality issues *stress: low-SNR and non-sinusoidal*

Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC/WECC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2022; NERC/WECC CA BESS 2023; Dong et al. Kaua'i 2021 analysis

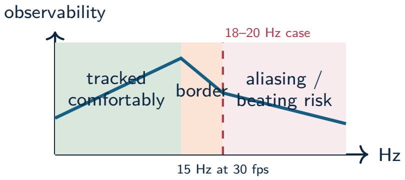
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Technical contributions

Scope boundary

not PMU certification

not a universal leaderboard

not hardware latency testing

1. OpenFreqBench
reproducible stress-test platform

2. Metric-complete protocol
RMSE, peak error, RoCoF error, T_{trip}, settling, CPU

3. IBR failure evidence
standard cases miss composite IBR failures

4. RA-EKF variant tested
explicit $\dot{\omega}$ state and event-gated adaptation

Contribution logic: explain estimator ranking shifts by objective, disturbance regime, and compute budget.

Experiment scale

16
estimators
loop, window, model

34
scenarios
standard + IBR

10
families
ramp to multi-event

60
MC runs
per pair

coverage
$16 \times 34 = \mathbf{544}$
summaries

replication
$544 \times 60 = \mathbf{32{,}640}$
records

sampling
1 MHz truth
10 kHz output

Experimental contract: fixed truth signals, scenarios, tuning policy, metrics, and timing before rankings are computed.

Central finding: no estimator dominates; winners change with event family, metric target, and compute budget.

Source: SGSMA benchmark matrix, Sec. III-IV and Fig. 2 caption

Estimator selection logic

Event regime
what stress dominates?

Ramp / RoCoF
phase jump
harmonics
multi-event

Estimator selection logic

Event regime
what stress dominates?

→

Objective metric
what failure matters?

Ramp / RoCoF
phase jump
harmonics
multi-event

RMSE
RoCoF error
T_{trip}
settling

Estimator selection logic

Event regime
what stress dominates?

Objective metric
what failure matters?

Compute envelope
what can run online?

Ramp / RoCoF
phase jump
harmonics
multi-event

RMSE
RoCoF error
T_{trip}
settling

MCU/DSP
FPGA
CPU/GPU
latency budget

Estimator selection logic

Event regime
what stress dominates?

→ **Objective metric**
what failure matters?

→ **Compute envelope**
what can run online?

→ **Evidence level**
replicated or pilot?

Event regime	Objective metric	Compute envelope	Evidence level
Ramp / RoCoF phase jump harmonics multi-event	RMSE RoCoF error T_{trip} settling	MCU/DSP FPGA CPU/GPU latency budget	SGSMA matrix (n=60) OpenFreqBench ATLAS severity validation gate

Estimator selection logic

Event regime
what stress dominates?

- Ramp / RoCoF
- phase jump
- harmonics
- multi-event

Objective metric
what failure matters?

- RMSE
- RoCoF error
- T_{trip}
- settling

Compute envelope
what can run online?

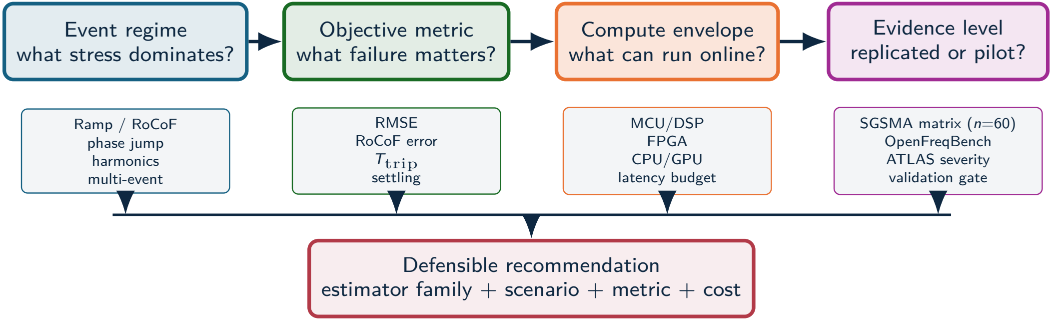
- MCU/DSP
- FPGA
- CPU/GPU
- latency budget

Evidence level
replicated or pilot?

- SGSMA matrix (n=60)
- OpenFreqBench
- ATLAS severity
- validation gate

Defensible recommendation
estimator family + scenario + metric + cost

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Benchmark & platform

What is missing in prior work — and OpenFreqBench

Benchmark literature gap

Standards
PMU compliance, metrology, calibration

Field data
real events, PMU/DFR streams

Open platforms
OpenPMU, openPDC, event libraries

Single-method studies
detailed but narrow comparisons

Missing layer
many estimators, same truth, same metrics, same policy

Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.

known truth → IBR stress → trip risk → CPU / latency

Estimator-level benchmark: one artifact contract for algorithms, scenarios, seeds, metrics, and plots

Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

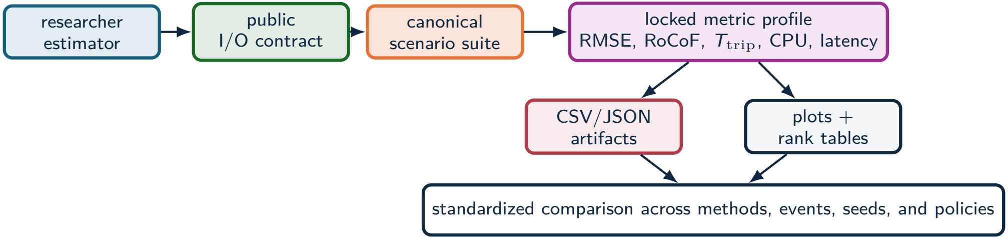
Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

Benchmark reports, metric tables, plots, manifests, and readiness gates.

Principle: a submitted estimator becomes a repeatable benchmark entry under the same scenarios, seeds, metrics, and artifacts.

Experimental method

Same truth, same metrics, fair tuning

Experimental method

truth synthesis
1 MHz waveform
known $f(t)$, RoCoF

relay stream
10 kHz samples
local voltage only

execution
estimator API
declared tuning grid

scoring
RMSE, peak, T_{trip}
settling, CPU

evidence
tables, traces
Pareto fronts

controlled
scenario script, truth, sampling rate, seeds

input contract
single-phase voltage; no PMU time-sync assumption

fair search
each estimator–scenario pair tuned inside its declared grid

multi-objective
accuracy, recovery, protection exposure, and compute kept separate

decision use
standard compliance, IBR stress, and deployment cost compared together

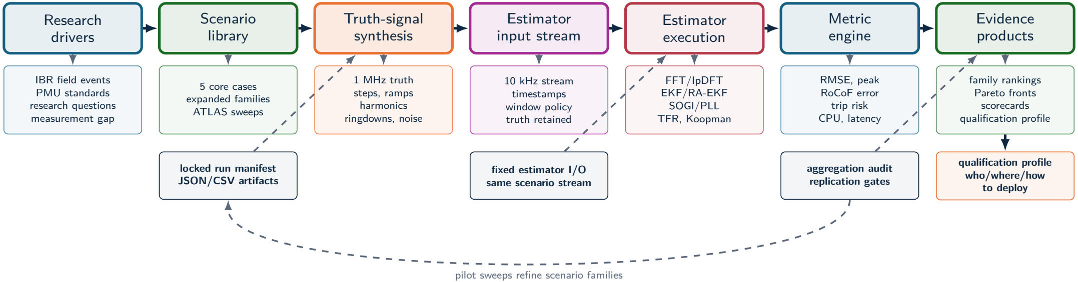
measurement scope
relay-class dynamic tracking

comparison rule
same truth, metrics, and timing

output rule
rank by scenario × metric × cost

Source: `Methodology and test framework`

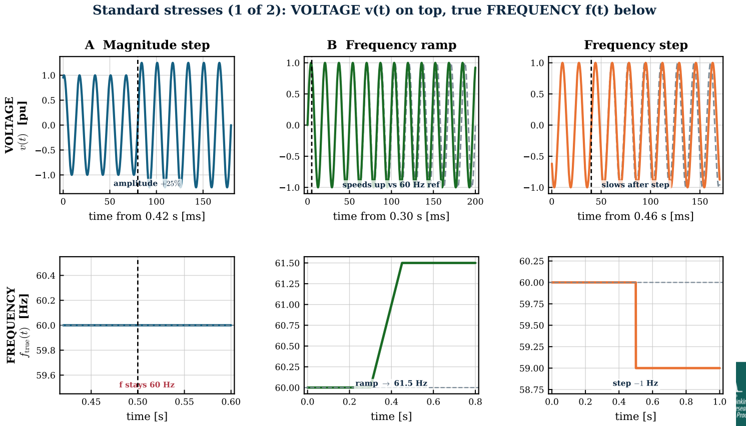
Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

Standard stresses (1 of 2): magnitude, ramp, frequency step

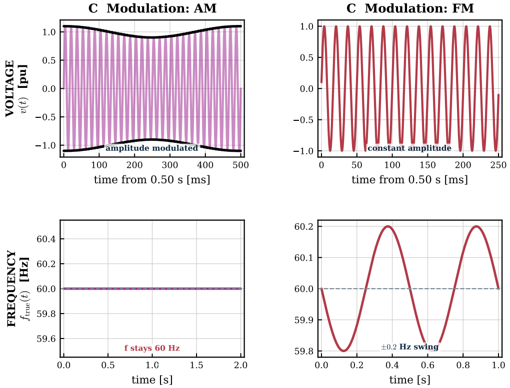
CORE SCENARIO SUITE



Standard stresses (2 of 2): modulation (AM and FM)

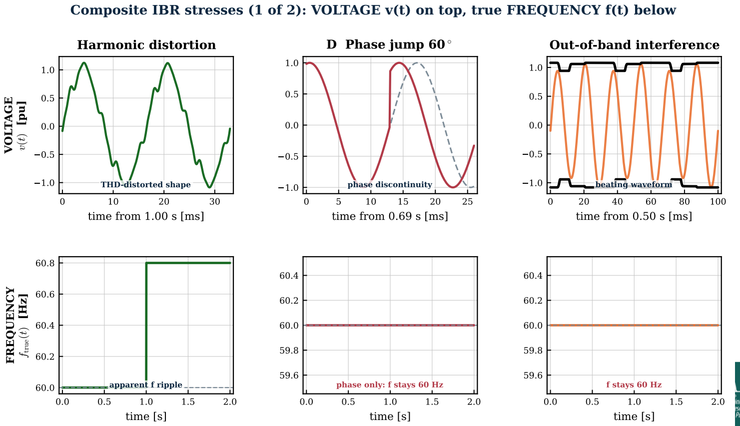
CORE SCENARIO SUITE

Standard stresses (2 of 2): VOLTAGE v(t) on top, true FREQUENCY f(t) below



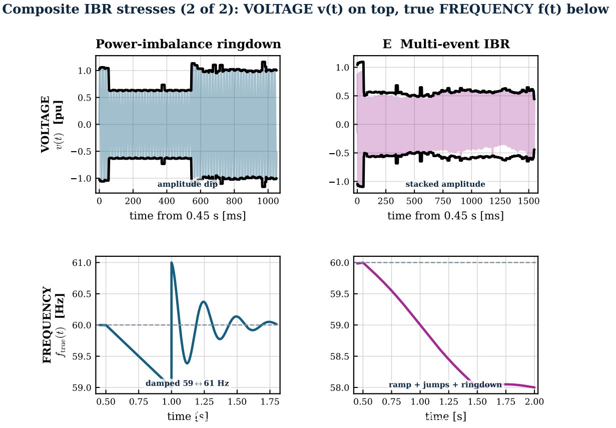
Composite IBR stresses (1 of 2): harmonics, phase jump, out-of-band

IBR STRESS SUITE



Composite IBR stresses (2 of 2): ringdown and multi-event

IBR STRESS SUITE



Scenario families: distortion, leakage, and low SNR

Compact signal model

$$v(t) = A(t)\sin\theta(t) + v_h(t) + v_x(t) + n(t)$$

$$v_h(t) = \sum_{h=2}^{H} a_h \sin(h\theta + \phi_h), \quad v_x(t) = \sum_{q} b_q \sin(2\pi f_q t + \psi_q)$$

$$\mathrm{THD} = \frac{\sqrt{\sum_{h\geq 2} V_h^2}}{V_1}, \quad \mathrm{SNR}_{dB} = 20\log_{10}\left(\frac{1/\sqrt{2}}{\sigma_n}\right)$$

Interpretation

The benchmark keeps the truth signal and the waveform perturbation separate. When $f_{\mathrm{true}}(t)$ is fixed, estimator error measures artifact rejection, not a physical frequency event.

Family	What it tests
Harmonics	Integer multiples of 60 Hz; THD rejection with fixed true frequency.
Interharmonics	Non-integer tones; leakage and beating that can mimic dynamics.
OOB tones	Out-of-band interference; filter selectivity, aliasing margins, PLL leakage.
Noise / SNR	Controlled σ_n; low-SNR crossings and covariance assumptions.

Why this slide matters

In these cases $f_{\mathrm{true}}(t)$ can remain nominal. Any frequency excursion reported by the estimator is then a waveform-induced measurement artifact.

Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL.

Window / spectral

IpDFT, TFT, ESPRIT, Prony, MUSIC.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF, LKF2.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Legacy baselines

TKEO and ZCD are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t:\ |e[\tau]| \le \epsilon,\ \forall \tau \ge t\}$$

$$C_{\mathrm{CPU}} = \mathrm{mean\ time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Core evidence

From standard cases to composite IBR stress

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**

RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	**0.006**	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	**0.028**
PLL	0.033	**1.47**	**0.227**
IpDFT	0.000	**14.4**	0.114
EKF	0.000	**37.1**	0.064

RMSE [Hz], mean over 5 seeds. Green: below 0.05 Hz guide; red: failure/divergence regime.

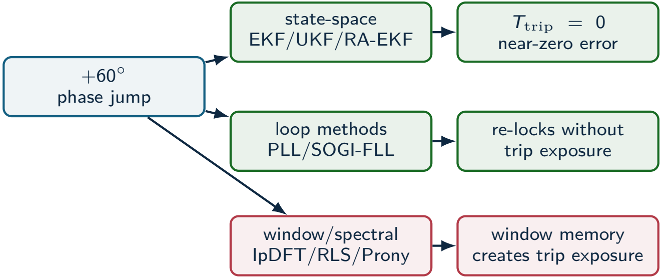
What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

A clean phase jump is a family separator

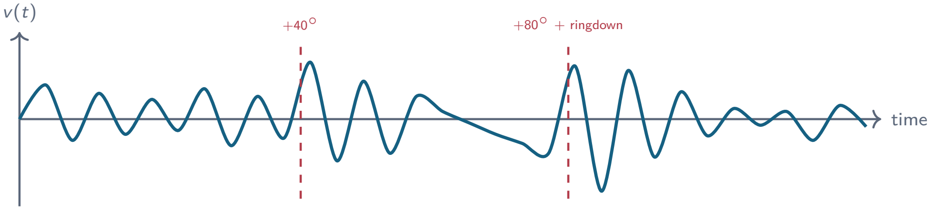


EKF, UKF, RA-EKF:
RMSE $\approx$ 0 Hz
$T_{\mathrm{trip}} = 0$

IpDFT: 1.56 Hz, 1.21 s
RLS: 2.28 Hz, 1.30 s
Prony: 2.39 Hz, 1.35 s

The clean jump does not expose RA-EKF's main advantage. It shows which estimator families carry memory through discontinuities.

Composite IBR sequence: why it is hardest



5th/7th harmonics → phase jump → −6 Hz/s RoCoF → ringdown recovery → impulsive noise

Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
UKF	**1.09**	**0.78**	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	**0.78**	11
SOGI-FLL	1.14	0.84	**9**
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	**15 241**
ZCD	**426**	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to $\approx$ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

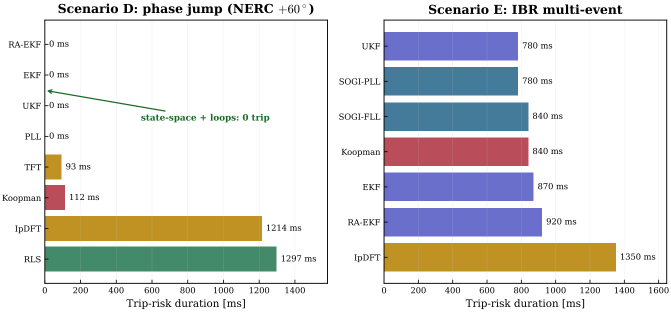
Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is 1000$\times$ costlier; RA-EKF sits mid-pack but never diverges.

Source: `full_mc_benchmark`, $n = 5$: `IBR_Multi_Event`

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric



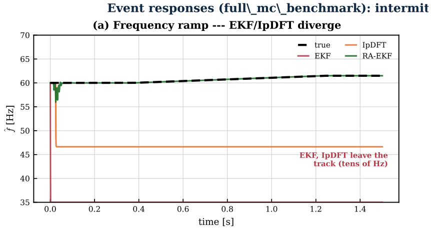
Result reading

- Scenario D: state-space filters and PLL keep $T_{\mathrm{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: *every* method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

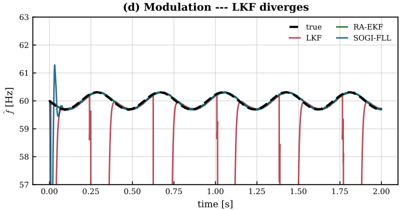
Result: T_{trip} separates protection exposure from average error.

The failures are visible in the trace, not only in the table

RAMP DIVERGENCE



MODULATION DIVERGENCE



Failure signature

EKF and IpDFT leave the true ramp almost immediately; after the event, the estimate is no longer operationally usable.

Failure signature

LKF stays plausible between collapses, then repeatedly drops out of the usable frequency band.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step $\approx$ 0; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\mathrm{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all $\approx$ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000$\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 μs/sample; Koopman $\sim$15,000	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Family	Target
SOGI-FLL	0.0348 μs	Loop	MCU
SOGI-PLL	0.0867 μs	Loop	MCU
EKF	0.813 μs	Model	MCU/DSP
RA-EKF	**1.61 μs**	**Model**	**DSP**
TFT / IpDFT	3–4 μs	Window	DSP/FPGA
Koopman	390 μs	Data-driven	CPU/GPU
ESPRIT	2,838 μs	Window	CPU

Source: SGSMA benchmark matrix, Sec. IV-D and Fig. 2(g)-(h)

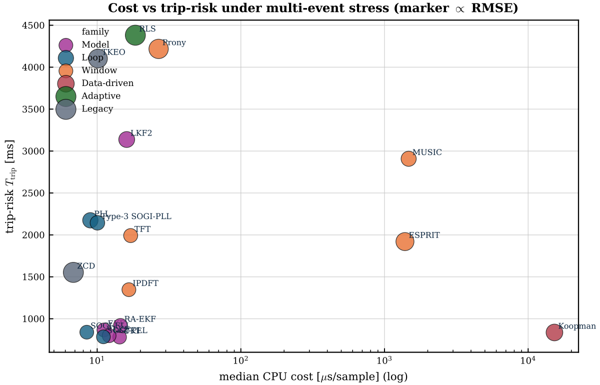
Deployment interpretation

CPU is a software-cost indicator from the benchmark harness, not a hardware-latency claim.

Key comparison

The practical low-cost cluster is SOGI-FLL, SOGI-PLL, EKF, and RA-EKF. Koopman and ESPRIT sit far to the right on cost without a global RMSE gain.

Deployment limits



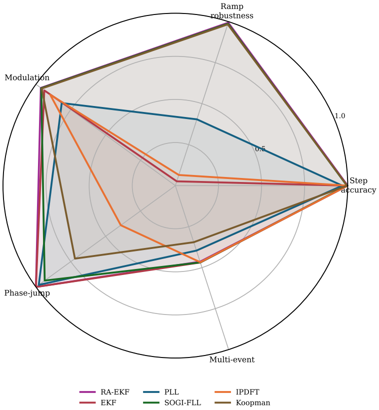
Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- In the OpenFreqBench expansion, state-space + loop methods cluster at low cost ($\leq 17\,\mu$s) and low trip.
- Spectral/data-driven methods trade lower error for much higher CPU in this artifact.
- RA-EKF stays low-cost with no ramp divergence in this run.

Source: `OpenFreqBench expansion: artifacts/full_mc_benchmark`, $n = 5$

Balance profile

Robustness profile --- $1/(1 + \mathrm{RMSE})$ per scenario (larger is better)



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + \mathrm{RMSE})$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P		
TFT	P	P	P		
IpDFT	P	F	P		
SOGI-FLL	P	P	P		
PLL	P	F	F*		
EKF	P	F	P		
UKF	P	P	P		
RA-EKF	P	P	P		
LKF2	F*	P	P		

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/T_{trip}/settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode. On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\mathrm{trip}} = 0$.

Metric disagreement

RMSE, trip-risk, and CPU select different winners. In the OpenFreqBench multi-event run, UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000$\times$ costlier.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved. The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility changes the accuracy ranking. In that same expansion, RA-EKF runs at $\sim$15 μs/sample; selected spectral and data-driven methods move far outside that cost range.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Source: `OpenFreqBench expansion: artifacts/full_mc_benchmark`, $n = 5$

Expanded & ATLAS

Does the pattern scale across families and severity?

Expanded stress analysis

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

SGSMA benchmark: family winners

Family	Count	RMSE leader	RMSE mean	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 ± 6.4 mHz	0 s (multiple)
Magnitude steps	7	EKF	0.388 ± 0.143 mHz	0 s (multiple)
Modulation	3	RA-EKF	17.8 ± 21.4 mHz	0 s (multiple)
Phase jumps	3	SOGI-FLL	174 ± 67.5 mHz	ZCD: 0.0194 s
OOB interference	1	EKF	13.6 ± 11.1 mHz	EKF / RA-EKF: 0 s
IBR harmonics	3	EKF	70.5 ± 27.3 mHz	TFT / Prony: 0.00947 s
IBR ringdown	5	UKF	313 ± 23.2 mHz	ESPRIT: 0.0989 s
IBR multi-event	1	PLL	173.5 ± 66.7 mHz	IpDFT: 0.0439 s

The paper result is conditional: SOGI-PLL leads ramps, EKF leads magnitude steps/OOB/harmonics, RA-EKF leads modulation, UKF leads ringdown, and PLL leads the composite IBR case. Trip-risk often points elsewhere.

Source: `SGSMA benchmark matrix, Table II`

ATLAS severity analysis

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Core cases	ATLAS severity curves	sign asymmetry	fixed vs oracle policy	validation gate
A–E failure mode	When does error start to grow?	Are positive and negative events symmetric?	Does tuning policy change the rank?	Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: `docs/ATLAS_PIPELINE.md`; `docs/ATLAS_METHOD_AUDIT.md`

ATLAS result: metric choice changes the answer

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Metric objective	Unique metric winner	Different from RMSE
Trip-risk exposure	26 / 113	5 / 26
RoCoF max error	112 / 113	53 / 112
RoCoF RMS error	112 / 113	48 / 112

Tie handling

Trip-risk has tied zero-exposure minima in 87/113 levels. It is an exposure screen first, not a single-winner leaderboard.

Why this matters

For protection-adjacent use, low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

RoCoF metrics change the preferred estimator in nearly half the severity map; trip-risk removes estimators with long exposure.

Source: `atlas-papergrade-missing-v1/global_metrics_report.csv`

ATLAS result: where estimators stop being valid

ATLAS VALIDATED SWEEP: $n = 30$**, fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

AM, FM, and interharmonics split the story

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	SOGI-FLL, TFT, and Koopman split the low-rate points; ESPRIT leads from 1–10 Hz.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF leads 0.5–15%; PLL leads at 20%.	EKF crosses the 0.05 Hz guide between 5% and 8%; most families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

When the textbook filter diverges

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

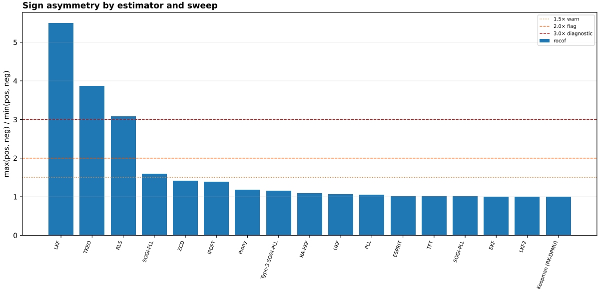
The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

RoCoF has a sign: several methods are down-ramp sensitive

ATLAS VALIDATED SWEEP: $n = 30$, fixed policy, both signs — readiness gate passed (9/9 sweeps)



Directional bias ($RMSE_-/RMSE_+$ at ± 50 Hz/s)

- LKF **5.5**× worse on −RoCoF
- TKEO **3.9**×, RLS **3.1**×
- SOGI-FLL 1.6×
- EKF, UKF, RA-EKF ≈1.0 (symmetric)

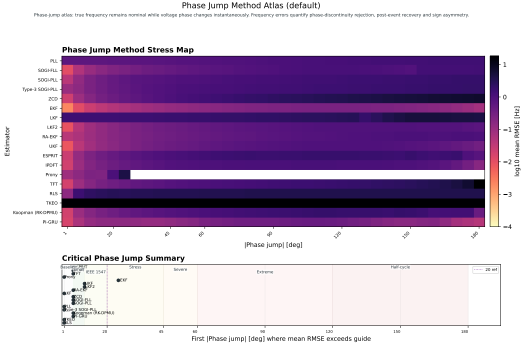
Why it matters

For LKF, TKEO, RLS, and SOGI-FLL, a +RoCoF-only test can certify behavior that does not hold on frequency *decline*. Both signs must be swept.

Source: `atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv` ($n = 30$, both signs)

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only **0.25 Hz @ 115°**, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps **2.5 → 21.9 Hz** between 150° and 180° — a phase-reversal singularity.

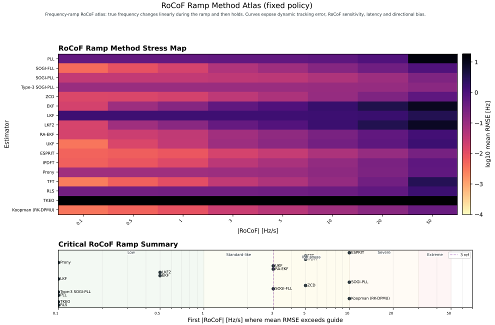
Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

Source: `atlas-phase-jump-1-180deg-5deg-all18/global_metrics_report.csv` (diagnostic, $n = 1$, default policy, positive sign)

No estimator owns the whole stress space

ATLAS VALIDATED SWEEP: $n = 30$**, fixed policy, both signs — readiness gate passed (9/9 sweeps)**

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	5.5 mHz
Standard (1–3)	ESPRIT	12.2 mHz
IBR stress (3–10)	ESPRIT	33.0 mHz
Severe (10–30)	SOGI-PLL	70.3 mHz



Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

The deliverable is a selection map: event class × metric × compute budget.

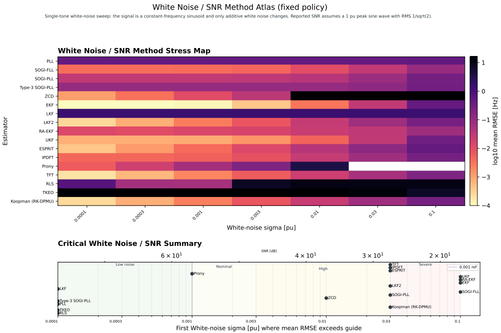
Source: atlas-papergrade-missing-v1/global_metrics_report.csv; RoCoF regimes averaged across both signs

ZCD is a narrow instrument, not a general estimator

ATLAS VALIDATED SWEEP: $n = 30$**, fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Low-stress champion

Up to ∼15% THD, zero-crossing detection gives the *lowest* error of any method — **0.27 mHz** at 1% THD, 2.2 mHz at 15%; and just 1.1 mHz at low broadband noise.



Expanded & ATLAS

ZCD is a narrow instrument, not a general estimator

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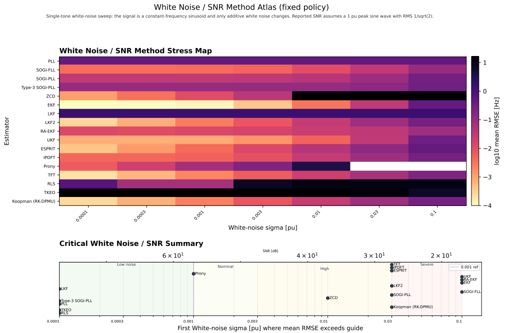
Broadband-noise cliff

Same method: 34 mHz at $\sigma = 0.003$ pu $\rightarrow$ **3112 Hz** at $\sigma = 0.1$ pu. EKF stays ≤ 0.43 Hz across the same sweep.



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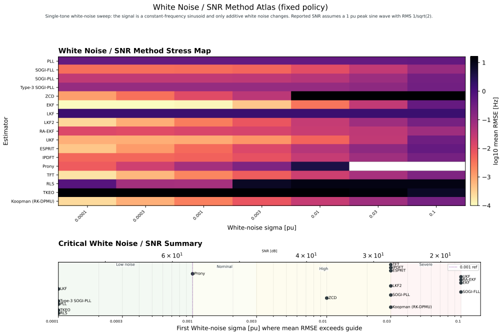
Harmonic cliff

Same method again: 2.2 mHz at 15% THD $\rightarrow$ **39 Hz** at 20% $\rightarrow$ **86 Hz** at 30%.



Voltage sags and swells are not symmetric

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n/(A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: `atlas-papergrade-missing-v1/global_metrics_report.csv`; `src/estimators/zcd.py`

What to claim

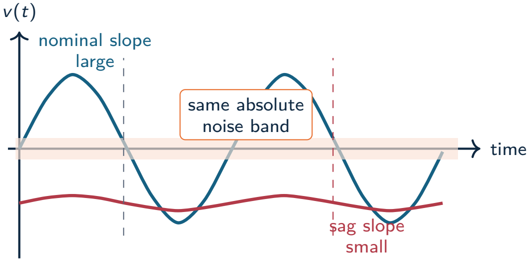
ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Why a voltage sag breaks zero-crossing detection

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left.\frac{dv}{dt}\right|_{t_0}} \qquad \left.\frac{dv}{dt}\right|_0 \approx A\omega$$

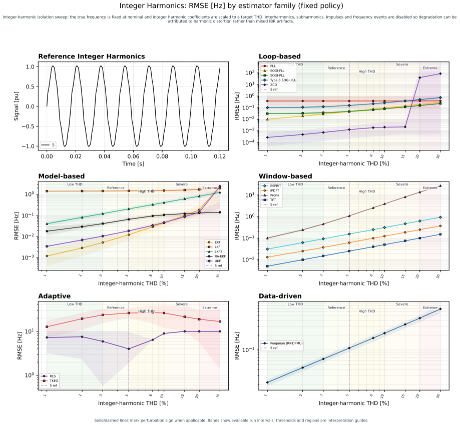
When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Harmonic degradation differs by estimator family

ATLAS VALIDATED SWEEP: $n = 30$, **fixed policy, both signs — readiness gate passed (9/9 sweeps)**

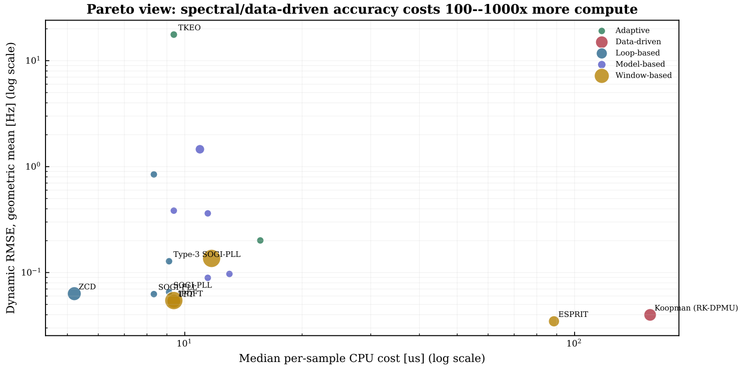


Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above $\sim$15%; window methods (TFT) degrade gently and lead at extreme THD.

Accuracy costs 100–1000× more compute

OPENFREQBENCH EXPANSION: 18 estimators, 33 scenarios, $n = 5$ seeds per scenario



Source: OpenFreqBench expansion: `artifacts/full_mc_benchmark`, $n = 5$

Low-cost core

PLL, SOGI-FLL, EKF, and RA-EKF sit in the deployable compute range.

Accuracy premium

Spectral and data-driven methods can move right by 100–1000× CPU cost.

Findings & deployment

What to deploy — and what comes next

Three findings

1. Metric-conditioned
The SGSMA matrix and ATLAS both show that RMSE, trip exposure, RoCoF, and CPU can select different methods.

2. Stress-bounded
AM remains survivable for many methods; severe FM, RoCoF, interharmonics, noise, and extreme THD break the 0.05 Hz guide.

3. Robustness over point accuracy
The OpenFreqBench ramp seeds expose silent divergence; RA-EKF removes that failure mode in the tested run.

Dynamic frequency estimators should be qualified by event family, sign, severity, recovery, trip exposure, and compute envelope.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Recommendation map

FFR / RoCoF tracking

↓

EKF / RA-EKF explicit $\dot{\omega}$

Recommendation map

FFR / RoCoF tracking → EKF / RA-EKF explicit $\dot{\omega}$

Anti-islanding phase jumps → PLL / RA-EKF near-zero T_{trip}

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Low-cost relay core → PLL / EKF CPU efficient

Recommendation map

- FFR / RoCoF tracking → EKF / RA-EKF explicit $\dot{\omega}$
- Anti-islanding phase jumps → PLL / RA-EKF near-zero T_{trip}
- Low-cost relay core → PLL / EKF CPU efficient
- Metering-like steady state → IpDFT / TFT if delay acceptable

Recommendation map

FFR / RoCoF tracking	Anti-islanding phase jumps	Low-cost relay core	Metering-like steady state
↓	↓	↓	↓
EKF / RA-EKF explicit $\dot{\omega}$	PLL / RA-EKF near-zero T_{trip}	PLL / EKF CPU efficient	IpDFT / TFT if delay acceptable

Deployability depends on event class, trip-risk tolerance, and compute budget.

Qualification profile

Core benchmark → Conditional estimator ranking → Metric-weighted event score → Dynamic estimator benchmark profile

Known truth signals → Estimator I/O contract → Locked metric registry → Machine-readable evidence

Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \, \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

Score composition

$\tilde{E}_s, \tilde{P}_s$	dynamic and peak error
$\tilde{T}_s, \tilde{S}_s$	trip exposure and recovery
$\tilde{C}_s, \tilde{L}_s$	compute and latency penalties

Why a score is needed

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score makes the deployment objective explicit.

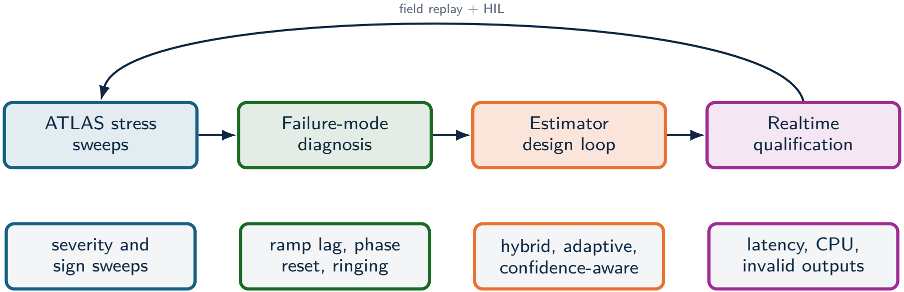
Protection profile

Raise γ, δ, λ when trip exposure, recovery, and latency dominate the risk model.

Monitoring profile

Raise α, β, η when reconstruction accuracy and compute budget dominate.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.